

DB Project



Team 10

- **Name :** Ahmed Riyad Mohamed Riyad
- **Name :** Abdelrahman Ibrahim Elias

Electronic system for examination and evaluation system

→ Main Entities :

Student
Professor
Course
Department
Question
Exam

→ Relations :

Department (1) --- (M) Course
Department (1) --- (M) Professor
Professor (M) --- (M) Course --> Course_Professor(teach)
Student (M) --- (M) Course --> Student_Course(register)
Exam (M) --- (M) Question --> Exam_Questions
Course (1) --- (M) Question
Exam (N) --- (M) Student --> Student_Exam(Result)
Student + Exam + Question --> إجابة كل طالب لكل سؤال في كل امتحان
Student_Exam (1) --- (N) Exam_Answer --- (1) Question

Student (M) --- (M) Course --> Course_Evaluation(evaluate)
Student + Professor + course --> Professor_Evaluation(evaluate)
Student (1) --- (M) Professor_Evaluation --- Course (1) --- (M)
Professor_Evaluation --- Professor (1) --- (M) Professor_Evaluation

Electronic system for examination and evaluation system

→ Q1 (Write a query that enables the students to view their results per course)

SELECT

**s.student_id,
CONCAT(s.first_name, ' ', s.last_name) AS student_name,
c.course_id,
c.name AS course_name,
sc.final_grade**

FROM Student s

JOIN Student_Course sc ON s.student_id = sc.student_id

JOIN Course c ON sc.course_id = c.course_id

Zagazig
Branch

Result

	student_id	student_name	course_id	course_name	final_grade
▶	1	Ahmed Riyad	1	Algorithms	85
	1	Ahmed Riyad	2	Databases	90
	2	Sara Mostafa	1	Algorithms	88
	3	Youssef Hany	3	Power Systems	79
	4	Laila Adel	4	Mechanics	91
	5	Omar Saeed	5	Concrete Design	74

Electronic system for examination and evaluation system

→ Q2 (Write a query that enables the head of department to see evaluation of each course and professor)

SELECT

```
d.name AS department_name,
c.course_id,
c.name AS course_name,
p.professor_id,
CONCAT(p.first_name, ' ', p.last_name) AS professor_name,
(SELECT AVG(rating)
FROM Course_Evaluation ce
WHERE ce.course_id = c.course_id) AS avg_course_rating,
(SELECT AVG(rating)
FROM Professor_Evaluation pe
WHERE pe.course_id = c.course_id
AND pe.professor_id = p.professor_id) AS avg_professor_rating
FROM Department d
JOIN Course c ON d.department_id = c.department_id
LEFT JOIN Professor p ON p.department_id = d.department_id
```

Result

	department_name	course_id	course_name	professor_id	professor_name	avg_course_rating	avg_professor_rating
▶	Computer Engineering	1	Algorithms	1	Ayman Khaled	4.5000	4.5000
	Computer Engineering	1	Algorithms	2	Rana Samir	4.5000	NULL
	Computer Engineering	2	Databases	1	Ayman Khaled	4.0000	NULL
	Computer Engineering	2	Databases	2	Rana Samir	4.0000	4.0000
	Electrical Engineering	3	Power Systems	3	Hossam Omar	5.0000	5.0000
	Electrical Engineering	3	Power Systems	4	Mona Fathy	5.0000	NULL
	Mechanical Engineering	4	Mechanics	5	Yasser Lotfy	3.0000	3.0000
	Mechanical Engineering	4	Mechanics	6	Nadine Maher	3.0000	NULL
	Civil Engineering	5	Concrete Design	7	Omar Hisham	NULL	NULL
	Civil Engineering	5	Concrete Design	8	Nour Gad	NULL	NULL
	Biomedical Engineering	6	Bioelectronics	9	Karim Refaat	NULL	NULL

Electronic system for examination and evaluation system

→ Q3 (Write a query that enables you to get top 10 high scores per course)

SELECT

**c.course_id,
c.name AS course_name,
s.student_id,
CONCAT(s.first_name, ' ', s.last_name) AS student_name,
sc.final_grade**

FROM Student_Course sc

JOIN Student s ON sc.student_id = s.student_id

JOIN Course c ON sc.course_id = c.course_id

ORDER BY c.course_id, sc.final_grade DESC

LIMIT 10;

Result

	course_id	course_name	student_id	student_name	final_grade
▶	1	Algorithms	2	Sara Mostafa	88
	1	Algorithms	1	Ahmed Riyad	85
	2	Databases	1	Ahmed Riyad	90
	3	Power Systems	3	Youssef Hany	79
	4	Mechanics	4	Laila Adel	91
	5	Concrete Design	5	Omar Saeed	74

Electronic system for examination and evaluation system

→ Q4 (Write a query to get the highest evaluation professor from the set of professors teaching the same course)

SELECT

**c.course_id,
c.name AS course_name,
p.professor_id,
CONCAT(p.first_name, ' ', p.last_name) AS professor_name,
AVG(pe.rating) AS avg_rating**

**FROM Professor_Evaluation pe
JOIN Professor p ON p.professor_id = pe.professor_id
JOIN Course c ON c.course_id = pe.course_id
GROUP BY c.course_id, p.professor_id
HAVING AVG(pe.rating) = (
SELECT MAX(avg_rating)
FROM (
SELECT AVG(rating) AS avg_rating
FROM Professor_Evaluation pe2
WHERE pe2.course_id = c.course_id
GROUP BY pe2.professor_id
) t
);**

Result

	course_id	course_name	professor_id	professor_name	avg_rating
▶	1	Algorithms	1	Ayman Khaled	4.5000
	2	Databases	2	Rana Samir	4.0000
	3	Power Systems	3	Hossam Omar	5.0000
	4	Mechanics	5	Yasser Lotfy	3.0000